

Manish Man Shrestha

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PROFILE SUMMARY

Embedded Firmware Engineer with 15 years of experience across industry, research, and embedded systems development, designing real-time firmware and low-level software for microcontroller-based and connected IoT systems. Strong background in Embedded C/C++, ARM Cortex-M, STM32, ESP32, bare-metal and FreeRTOS-based firmware, task scheduling, concurrency, interrupts, device drivers, and hardware-software integration. Experienced in secure device-to-cloud communication using MQTT/MQTTs, TLS, Wi-Fi, and BLE, with hands-on work in encryption, authenticated telemetry, OTA firmware updates, and fault-tolerant embedded systems. Skilled in optimizing firmware for timing, memory usage, reliability, and real-time performance, with hardware debugging experience using JTAG/SWD, oscilloscopes, and logic analyzers.

CORE SKILLS

- **Embedded Firmware Development:** Embedded C/C++, real-time firmware, bare-metal programming, device drivers, board bring-up, interrupt-driven and event-driven systems, low-level hardware programming, firmware-hardware integration
- **RTOS & Real-Time Systems:** FreeRTOS, task scheduling, concurrency, inter-task communication, queues, semaphores, mutexes, timing optimization, watchdogs, real-time task management
- **Embedded Security:** TLS, AES, ChaCha20-Poly1305, MQTT over TLS, authenticated and encrypted telemetry, post-quantum cryptography (Kyber/ML-KEM, Dilithium/ML-DSA)
- **Connectivity & IoT Protocols:** TCP/IP, lwIP, MQTT/MQTTs, Wi-Fi, BLE, Modbus RTU, RS-232/485, CAN, BACnet
- **MCUs / SoCs / SBCs:** ARM Cortex-M, STM32, NXP/LPC, ESP32/ESP-IDF, Raspberry Pi, PIC, AVR, 8051
- **Hardware Interfaces:** UART, SPI, I2C, GPIO, ADC/DAC, PWM, DMA, USB CDC, memory-mapped I/O
- **Memory & Boot Systems:** EEPROM, NOR/NAND Flash, SPI Flash, NVS, bootloaders, OTA-safe firmware updates, firmware verification and rollback
- **Debugging & Bring-Up Tools:** JTAG/SWD debugging, oscilloscopes, logic analyzers, serial debugging, firmware trace analysis, CAN analyzer
- **Control & Electromechanical Systems:** ROS 2 / micro-ROS, PWM motor/control systems, ADC feedback, sensor integration, power management, real-time control logic
- **Signal Processing:** Embedded DSP, FFT, IIR/FIR filters, Butterworth filtering, MATLAB-based signal analysis, CMSIS-DSP
- **Programming Languages:** C, C++, Python, C#, R
- **Development Platforms & Tools:** ESP-IDF, Keil MDK, MPLAB, Visual Studio, Git, Altium Designer, Eagle, KiCad, OpenSCAD, Azure IoT Hub

WORK EXPERIENCE

Graduate Research Assistant

January 2023 - Present

South Dakota State University, Brookings, South Dakota

- Architected and developed Embedded C firmware on ESP32/ESP32-S3 using FreeRTOS for real-time sensor acquisition, device drivers, Modbus RTU communication, and secure MQTT/TLS telemetry over Wi-Fi.
- Designed and integrated MCU drivers for sensors and peripherals, enabling reliable data acquisition from soil moisture, temperature, humidity, nitrate, and chloride sensing systems.
- Performed hardware bring-up, peripheral integration, and low-level debugging for sensor-based embedded systems using SPI, I2C, UART, RS-485, GPIO, and ADC interfaces.
- Supported PCB design, schematic review, and board-level integration using KiCad for custom IoT sensing hardware.
- Engineered secure telemetry pipelines using AES, MQTT over TLS, Kyber, Dilithium, and ChaCha20-Poly1305, with backend validation and visualization using C#/.NET tools.
- Designed and validated embedded systems using structured testing, debugging, and verification workflows to ensure reliable operation in real-world greenhouse and hydroponic environments.
- Designed and 3D-printed custom OpenSCAD prototypes with integrated sensor mounts, leak-resistant fluid containment, drainage ports, and hose interfaces for leachate collection.

Assistant Professor

August 2017 - December 2022

Cosmos College Of Management and Technology, Lalitpur, Nepal

- Taught embedded systems, microcontroller programming, and real-time firmware development.
- Led embedded R&D projects involving MCU firmware, sensor integration, and hardware-software systems.

CTO / Technical Consultant

April 2017 - December 2022

Samadhan Engineering Pvt. Ltd., Lalitpur, Nepal

- Led embedded systems development from concept to deployment, including requirements review, firmware architecture, hardware integration, testing, documentation, and production validation.
- Developed real-time firmware using PWM, timers, ADC feedback, RTOS scheduling, sensor interfaces, and communication protocols on microcontroller platforms.
- Developed CAN bus driver code for an access management system involving motorized door control, and used a CAN analyzer to inspect, verify, and debug CAN message traffic during system integration.
- Designed and reviewed PCB layouts using Altium Designer for embedded sensing and control systems, supporting board bring-up, testing, and hardware-firmware debugging.

- Developed BLE-based telemetry for a structural health monitoring prototype to transmit damage-index data, later migrating the system to Wi-Fi for improved deployment reliability.
- Designed embedded firmware architectures for field-deployed systems with robust error handling, communication reliability, and maintainable driver-level code.
- Established Git-based version control and YAML-based CI/CD workflows using Docker images to generate repeatable firmware builds and manage release artifacts for embedded systems.

Research Scientist

March 2016 - March 2017

KAIST - Measurement Robotics & Diagnostic AI Lab, Daejeon, South Korea

- Developed embedded DSP firmware with lwIP TCP/IP networking for wireless structural monitoring systems.
- Implemented high-speed signal acquisition and wireless communication pipelines on ARM-based embedded platforms.
- Supported sensor electronics integration, signal validation, and embedded system testing for real-time structural monitoring applications.

Research Assistant

August 2013 - February 2016

Jeonbuk National University, Jeonju, South Korea

- Developed firmware for a dual-core ARM Cortex-M4/M0 LPC4370 system for ultrasonic signal generation, synchronized data acquisition, and real-time DSP processing.
- Configured the LPC4370's built-in USB peripheral as a virtual COM port to transfer ultrasonic waveform data to a laptop for verification before migrating wireless transmission to an external Bluegiga Wi-Fi modem.
- Designed and supported PCB layouts using Eagle for embedded ultrasonic sensing hardware and assisted with board bring-up and hardware debugging.
- Developed Qt-based GUI tools to control a laser excitation system, acquire ultrasonic wave data from an embedded sensing device, and visualize/analyze received waveforms for structural monitoring experiments.
- Implemented FFT, IIR, and FIR filtering algorithms for embedded structural monitoring applications.
- Supported hardware–software integration, signal validation, and debugging for real-time sensing systems.

Firmware Engineer

March 2011 - August 2013

Real Time Solutions Pvt. Ltd., Lalitpur, Nepal

- Developed embedded C/C++ firmware for ARM, PIC, and 8051 microcontrollers, including device drivers, communication stacks, and RTOS-based task scheduling.
- Implemented Protothreads-based cooperative multitasking on 8051 and PIC microcontrollers and developed custom bare-metal schedulers for resource-constrained embedded systems.
- Implemented communication protocols including UART, SPI, Modbus RTU, and cellular telemetry using AT commands.
- Developed firmware for industrial and field-deployed systems including panel meters, MPPT solar battery chargers, queue management systems, and flood early-warning devices.
- Debugged hardware–firmware interactions using oscilloscopes, logic analyzers, and serial debugging tools to improve system reliability.

EDUCATION		
•	Ph.D. , Agricultural & Biosystems Engineering - South Dakota State University, USA	Aug 2026
•	Graduate Certificate , Data Science - South Dakota State University, USA	Aug 2025
•	M.E. , Mechatronics Engineering - Jeonbuk National University, South Korea	Dec 2016
•	B.E. , Electronics & Communication - Cosmos College of Management and Technology, Nepal	Dec 2010

SELECTED PROJECT

- **Distributed IoT Monitoring System with Post-Quantum Security | ESP32-S3, FreeRTOS:** Developed real-time embedded firmware for a distributed IoT sensing system using ESP32-S3, FreeRTOS, sensor interfaces, secure MQTT/TLS device-to-cloud telemetry, encrypted data transmission, signature verification, post-quantum cryptography, sensor validation, fault handling, and backend integration.
- **Secure IoT Hydroponics Controller | ESP32-S3, FreeRTOS:** Developed FreeRTOS-based firmware integrating Modbus RTU nitrate/chloride sensors, environmental sensors, control outputs, MQTT/TLS telemetry, Kyber, Dilithium, ChaCha20-Poly1305 security, OTA firmware updates, sensor validation, communication recovery, and fault-handling mechanisms.
- **Multichannel Ultrasonic Structural Health Monitoring Device | STM32 ARM Cortex-M4:** Developed real-time STM32 firmware for high-speed multichannel signal acquisition, programmable ultrasonic excitation, synchronized actuation and sensing, interrupt-driven acquisition, deterministic timing, and embedded DSP including FFT and IIR filtering.
- **Flood Early-Warning Node | ARM Cortex-M0:** Developed embedded firmware for a remote ARM Cortex-M0 monitoring device integrating environmental sensors, signal conditioning, low-power operation, real-time data acquisition, digital filtering, communication recovery, fault handling, and reliable telemetry for continuous monitoring and remote diagnostics.

PUBLICATIONS (SELECTED)	
•	Structural Health Monitoring (SAGE): Wireless ultrasonic device for aircraft structures (2017).
•	Smart Agricultural Technology (ELSEVIER): Distributed IoT + PQC for nitrate/soil-moisture monitoring in corn production (2026).